

Nathan Zhao

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Education

Georgia Institute of Technology <i>Bachelor of Science in Electrical Engineering</i>	Atlanta, Georgia
The Pembroke Hill School <i>High School Diploma, Cum Laude</i>	Kansas City, Missouri
• Relevant Coursework: ACT: 36; GPA: 97.7; Multivariable Calculus, Differential Equations	

Work Experience

Research and Development Intern @ MRIGlobal	2025
• Designed and fabricated custom chemically coated gas sensor array and STM32 PCB (Altium), enabling real-time detection and classification of chemical warfare agents	
• Deployed CNN-LSTM model on-device using TensorFlow Lite for real-time edge inference, wrote embedded C++ firmware and Python GUI tools for data acquisition/processing	
Research Assistant @ University of Missouri-Kansas City, Multiplatform Interactive Robotics Lab (mirolab)	2025
• Worked under Professor Hee Sup Shin on vision-based finger goniometry with FDM-printed ArUco wearable	
• Developed Python programs for Kalman fusion and filtering of MediaPipe hand landmark detection with stereo 6DOF fiducial marker pose estimation	
Research Intern @ University of Pittsburgh, Accessible SmartTech Lab	2024
• Worked under Professor Dan Ding with ROS Noetic in Ubuntu 20.04 on a Hello Robot Stretch RE1	
• Programmed autonomous functionality using ArUco marker detection and 2D LiDAR SLAM with RViz	
Intern @ WeCode KC	2022
• Developed video game interactive demo to teach young students programming concepts using C# and Unity	

Selected Projects

Design and Development of a Bimanual Mobile Manipulator for Assisted Living	2024 - Present
• Onshape, C++, C#, PlatformIO, $\LaTeX$	
Unified Multi-View Kalman Fusion of ArUco Marker Pose and Vision-Based Hand Landmarks for Continuous Finger Joint Angle Estimation	2025 - 2026
• Python, OpenCV, MediaPipe, Onshape, $\LaTeX$	
mmHome: Development of a Multimodal IoT Network for Contactless Vital Sign Health Monitoring and Fall Detection	2025 - 2026
• Python, scikit-learn, C++, PlatformIO	
Multi-Agent Swarm Control with Real-Time Model-Predictive Control of Reynolds Boids Parameters	2026 - Present
• Python, PyBullet, PyTorch, $\LaTeX$	

Relevant Skills

Programming: C++, Python, C#, Java, MATLAB, $\LaTeX$, Git
Robotics/ML: OpenCV, PyTorch, scikit-learn, ROS, TensorFlow Lite, Linux (Ubuntu)
Design/Hardware: SolidWorks, Onshape, Altium Designer, KiCAD, Arduino, PlatformIO
Languages: English, Chinese (Mandarin), Spanish